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Seat device

Abstract

To better fulfill a sleep-inducing function for a seated person, a seat device includes a pressing device provided in a seat back and configured to press a back or a waist of a seated person to induce breathing; at least one of a temperature adjusting device configured to change a temperature of a seat cushion and/or the seat back and a shape adjusting device configured to change a surface shape of the seat cushion and/or the seat back; and a controller configured to control the pressing device to press the back or the waist of the seated person at a set cycle corresponding to a breathing cycle of a person at a sleeping time, and control the at least one of the temperature adjusting device and the shape adjusting device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is the U.S. National Stage entry of International Application No. PCT/JP2020/044876 filed under the Patent Cooperation Treaty on Dec. 2, 2020, which claims priority to U.S. Provisional Patent Application No. 62/942,466 filed on Dec. 2, 2019 and U.S. Provisional Patent Application No. 62/951,623 filed on Dec. 20, 2019, all of which are incorporated herein by reference.

TECHNICAL FIELD
(1) The present invention relates to a seat device, and more particularly, to a seat device that has a sleep-inducing function for a seated person.
BACKGROUND ART
(2) As a seat device for a vehicle such as an automobile, a seat device having a function of inducing breathing of a seated person is known (for example, Patent Documents 1 and 2). This seat device induces the breathing of a prescribed cycle that is suitable for relaxation by applying on a prescribed cycle a stimulus such as compression or pressure to a portion of a back of the seated person that corresponds to a lumbar vertebra or a thoracic vertebra.
PRIOR ART DOCUMENT(S)
Patent Document(s)
(3) Patent Document 1: JP2005-296452A Patent Document 2: JP2012-65728A
SUMMARY OF THE INVENTION
Task to be Accomplished by the Invention
(4) To enable a person traveling by a vehicle such as an automobile, an aircraft, and a ship to recover from a lack of sleep, it is useful to make an improvement for inducing sleep of the person seated on a seat device and letting the person sleep comfortably.
(5) In an automobile having an autonomous driving function that is being developed in recent years, the driver can sleep while driving the automobile. Accordingly, a seat device that fulfills a sleep-inducing function for the seated person becomes more useful for a driver's seat.
(6) For these reasons, there is an increasing demand for the development of the seat device that can better fulfill a sleep-inducing function for the seated person.
(7) In view of the above background, an object of the present invention is to provide a seat device that can better fulfill a sleep-inducing function for a seated person.
Means for Accomplishing the Task
(8) To achieve such an object, one aspect of the present invention provides a seat device,

comprising: a seat body (**35**) including a seat cushion (**30**) and a seat back (**32**); a pressing device (**40**, **42**) provided in the seat back and configured to press a back or a waist of a seated person so as to induce breathing thereof; at least one of a temperature adjusting device (**52**) and a shape adjusting device (**44**), the temperature adjusting device configured to change a temperature of the seat cushion and/or the seat back, the shape adjusting device configured to change a surface shape of the seat cushion and/or the seat back; and a controller (**100**) configured to control the pressing device such that the pressing device presses the back or the waist of the seated person at a set cycle corresponding to a breathing cycle of a person at a sleeping time, and control the at least one of the temperature adjusting device and the shape adjusting device.
(9) According to this aspect, it is possible to better fulfill a sleep-inducing function for the seated person.
(10) In the above aspect, preferably, the controller is configured to correct the set cycle upon learning the breathing cycle of the seated person at the sleeping time.
(11) According to this aspect, it is possible to properly induce sleep by inducing breathing.
(12) In the above aspect, preferably, the controller is configured to determine whether the breathing of the seated person is abdominal breathing or thoracic breathing, and control the pressing device so as to induce the breathing corresponding to a determination result thereof.
(13) According to this aspect, it is possible to properly induce the breathing regardless of whether the breathing of the seated person is the abdominal breathing or the thoracic breathing.
(14) In the above aspect, preferably, the controller is configured to control the pressing device so as to induce abdominal breathing when sleep is induced, and induce thoracic breathing when the sleep ends.
(15) According to this aspect, when sleep is induced, it is possible to induce the breathing suitable for inducing sleep. On the other hand, when sleep ends, that is, when the seated person wakes up, it is possible to induce the breathing suitable for waking up the seated person.
(16) In the above aspect, preferably, the seat device further comprises a breathing detecting device (**72**) configured to detect the breathing of the seated person, wherein in a case where the breathing detected by the breathing detecting device is unstable, the controller controls the pressing device so as to induce the breathing that eliminates unstableness in the breathing.
(17) According to this aspect, it is possible to promote the elimination of the unstableness in the breathing of the seated person.
(18) In the above aspect, preferably, the controller is configured to determine based on external information whether the seated person is likely to sleep, and turn on the pressing device and perform control to cause the temperature adjusting device to warm the seat cushion and the seat back upon determining that the seated person is likely to sleep.
(19) According to this aspect, when the seated person is likely to sleep, it is possible to better induce sleep of the seated person by inducing the breathing suitable for inducing sleep and warming the seat cushion and the seat back.
(20) In the above aspect, preferably, the seat device further comprises a warm air device (**58**) configured to warm feet of the seated person, wherein the controller is configured to determine based on external information whether the seated person is likely to sleep, and perform control to turn on the pressing device and the warm air device upon determining that the seated person is likely to sleep.
(21) According to this aspect, when the seated person is likely to sleep, it is possible to better induce sleep of the seated person by inducing the breathing suitable for inducing sleep and warming the feet of the seated person.
(22) In the above aspect, preferably, the controller is configured to control the temperature adjusting device such that temperatures of the seat cushion and the seat back rise and fall repeatedly.
(23) According to this aspect, it is possible to relax the muscle of the seated person and thus cause a

relaxing effect on the seated person.
(24) In the above aspect, preferably, the controller is configured to determine, based on external information, whether the seated person is likely to sleep, and turn on the pressing device and control the shape adjusting device such that a massage synchronized with a breathing rhythm of the seated person is given upon determining that the seated person is likely to sleep.
(25) According to this aspect, when the seated person is likely to sleep, it is possible to better induce sleep of the seated person by inducing the breathing suitable for inducing sleep and massaging the seated person.
(26) In the above aspect, preferably, when the seated person is likely to sleep, the controller is configured to control the temperature adjusting device such that temperatures of the seat cushion and the seat back rise.
(27) According to this aspect, it is possible to improve a massaging effect when the seated person is likely to sleep.
(28) In the above aspect, preferably, the controller is configured to control the shape adjusting device such that a massage synchronized with a breathing rhythm of the seated person is given.
(29) According to this aspect, it is possible to have a preferable massaging effect.
(30) In the above aspect, preferably, the seat device further comprises a body pressure detecting device (**46**) configured to individually detect a body pressure of the seated person in each portion of the seat cushion and the seat back, wherein the controller is configured to control the shape adjusting device such that a difference in the body pressure in each portion is reduced.
(31) According to this aspect, it is possible to distribute the body pressure and thus reduce the fatigue of the seated person seated for a long time.
(32) In the above aspect, preferably, the controller is configured to control the shape adjusting device such that surfaces of the seat cushion and the seat back are deformed so as to prompt the seated person to change a posture.
(33) According to this aspect, it is possible to prompt the seated person to roll over.
(34) In the above aspect, preferably, the controller is configured to control the shape adjusting device such that surfaces of the seat cushion and the seat back are deformed at a set wake-up time so as to promote the seated person to wake up.
(35) According to this aspect, it is possible to prompt the seated person to wake up at the wake-up time.
(36) In the above aspect, preferably, the seat device consists of a seat device for a vehicle, more specifically, a seat device for an automobile configured to be driven autonomously.
Effect of the Invention
(37) Thus, according to the above aspects, it is possible to provide a seat device that can better fulfill a sleep-inducing function for a seated person.

Description

BRIEF DESCRIPTION OF THE DRAWING(S)
(1) FIG. **1** is a perspective view showing a seat device according to an embodiment of the present invention;
(2) FIG. **2** is a block diagram of a control system of the seat device according to the present embodiment:
(3) FIG. **3** is a general flowthart of the control system of the seat device according to the present embodiment;
(4) FIG. **4** is a flowchart of a posture correction routine of the seat device according to the present embodiment;
(5) FIG. **5** is a flowchart of a warming/cooling routine of the seat device according to the present

embodiment;
(6) FIG. **6** is a flowchart of a massage routine of the seal device according to the present embodiment;
(7) FIG. **7** is a flowchart of a breathing induction routine of the seat device according to the present embodiment;
(8) FIG. **8** is a perspective view showing a structure of the seat device according to the present embodiment;
(9) FIG. **9** is an enlarged cross-sectional view showing a main portion of a power supply mechanism of the seat device according to the present embodiment;
(10) FIG. **10** is an enlarged cross-sectional view showing a main portion of a power supply mechanism of a seat device according to another embodiment;
(11) FIG. **11** is a cross-sectional view showing a main portion of a power supply mechanism of a seat device according to still another embodiment;
(12) FIG. **12** is an enlarged cross-sectional view showing a main portion of a power supply mechanism of a seat device according to still another embodiment;
(13) FIG. **13** is a perspective view showing a structure of a seat device according to still another embodiment;
(14) FIG. **14** is a cross-sectional view showing a main portion of a power supply mechanism in a seat device according to still another embodiment; and
(15) FIG. **15** is a plan view showing a main portion of a power supply mechanism of a seat device according to still another embodiment.
DETAILED DESCRIPTION OF THE INVENTION
(16) In the following, one embodiment in which a seat device according to the present invention is applied to an automobile having an autonomous driving function will be described with reference to the drawings.
(17) As shown in FIG. **1**, a seat device **10** according to the present embodiment is provided on a floor panel **14** of the automobile via left and right slide rail devices **12** extending in the front-and-rear direction. Each slide rail device **12** includes a lower rail **16** fixed to the floor panel **14** via brackets **15**, and an upper rail **18** (upper slider) fixed to a lower surface of a seat cushion **30** and engaged with the lower rail **16** so as to slide in the front-and-rear direction.
(18) A side door **20** is provided on a lateral side of the seat device **10**. A window pane **22** is attached to an upper portion of the side door **20**.
(19) The seat device **10** includes the seat cushion **30** forming a seat surface, and a seat back **34** forming a backrest surface and connected to a rear portion of the seat cushion **30** such that the seal back **34** can be inclined by an electric reclining device **32** (hereinafter abbreviated as “reclining device”). A headrest **36** is attached to an upper portion of the seat back **34**. An electric ottoman **38** (hereinafter abbreviated as “ottoman **38**”) is provided at a front portion of the seat cushion **30**. The ottoman **38** is rotatable between a shown rotation position in which the ottoman **38** extends substantially vertically and an unshown rotation position in which the ottoman **38** extends substantially horizontally. The ottoman **38** can be fixed at any rotation position.
(20) The seat back **34** is provided with a back pressing device **40** and a waist pressing device **42**. The back pressing device **40** and the waist pressing device **42** each consist of an air pad that expands and deforms as compressed air is supplied from a compressed air source (not shown) to the air pad. The back pressing device **40** protrudes forward from a front surface of the seal back **34** thereby pressing the back (more specifically, a portion corresponding to a thoracic vertebra around a scapula) of a seated person so as to induce thoracic breathing. The waist pressing device **42** protrudes forward from the front surface of the seat back **34**, thereby pressing the waist (more specifically, a portion corresponding to a lumbar vertebra) of the seated person to induce abdominal breathing.
(21) The seat cushion **30** and the seat back **34** are each provided with seat shape adjusting devices

44 at positions excluding the back pressing device **40** and the waist pressing device **42**. The seat shape adjusting devices **44** change surface shapes of the seat cushion **30** and the seat back **34**. Each seat shape adjusting device **44** consists of an air pad that expands and deforms as compressed air is supplied from a compressed air source (not shown) to the air pad. Each seat shape adjusting device **44** deforms the upper surface of the seat cushion **30** upward or the front surface of the seat back **34** forward as the air pad expands and deforms.

(22) Each seat shape adjusting device **44** massages the thigh and back of the seated person and corrects the body pressure distribution thereof by protruding the seat cushion **30** and the front surface of the seat back **34**. The seat shape adjusting devices **44** are combined with each other, and thus change the surface shapes of the seat cushion **30** and the seat back **34**, thereby correcting the seated posture of the seated person and prompting the seated person to roll over while sleeping. The seat shape adjusting devices **44** move with respect to each other, and thus also function as an exercise device.

(23) The seat cushion **30** and the seat back **34** are each provided with body pressure detecting devices **46** in the vicinity of each seat shape adjusting device **44**. The body pressure detecting devices **46** individually detect the body pressure of the seated person.

(24) An electric neck massaging device **48** that massages the neck of the seated person is installed in the headrest **36**. The neck massaging device **48** locally presses the neck so as to press acupoints called “Fuchi”, “Tenchu”, and “Kankotsu”. Pressings these three acupoints is said to be effective in relieving eyestrain, a stiff shoulder, and a stiff neck, and thus has the effect of reducing fatigue of the seated person.

(25) Electric calf massaging devices **50** that massage the calves of the seated person are installed in the ottoman **38**. Each calf massaging device **50** locally presses the calf so as to press acupoints called “Shokin”, “Shokan”, “Shozan”, “chikuhin”, and “Hiyou”. Pressing these five acupoints is said to be effective in reducing numbness, improving blood circulation, and relieving swelling and fatigue, and thus has the effect of reducing fatigue of the seated person.

(26) The seat cushion **30** and the seat back **34** are provided with seat temperature adjusting devices **52** that change temperatures of the seat cushion **30** and the seat back **34**. Each seat temperature adjusting device **52** consists of an electric heater for raising the temperature and a Peltier element or a cooling air supply device such as an air ventilation system (AVS) for lowering the temperature. The seat temperature adjusting devices **52** adjust surface temperatures of the seat cushion **30** and the seat back **34** within a prescribed temperature range, for example, 24° C. to 36° C. Incidentally, the ottoman **38** may also be provided with the seat temperature adjusting device **52**.

(27) On a front side of the seat cushion **30**, knee back heaters **56** are provided at portions with which knee backs (backs of knees) of the seated person come into contact. Each knee back heater **56** directly warms the knee back of the seated person. A fat layer of the knee back is thin and an aorta passes through the knee back, which makes the knee back sensitive to stimuli. Accordingly, as the knee back heater **56** warms the knee back, the warmth is directly transmitted to the body, so that a thermal effect can be enhanced and fatigue can be effectively reduced.

(28) On a lower surface of the seat cushion **30**, a warm air device **58** that blows warm air toward the feet of the seated person is provided. The lower limbs including the feet are portions on which much blood of the whole body is concentrated. Accordingly, as the warm air blown by the warm air device **58** warms the feet, blood circulation can be improved, metabolism can be enhanced, oversensitivity to the cold can be relieved, and thus the health of the seated person can be improved.

(29) A blanket box **60** is attached to a side portion of the upper rail **18** on one of left and right sides. A blanket **62** is stored in the blanket box **60**. One end of the blanket **62** is fixed to the blanket box **60**. The other end of the blanket **62** is provided with engagement hooks **66** configured to removably engage with locking holes **64** provided on the upper rail **18** on the other of the left and right sides.

(30) The blanket **62** can be pulled out from the blanket box **60**. The blanket **62** functions as a lap

robe as a thermal member as the engagement hooks **66** engage with the locking holes **64** and thus the blanket **62** stretches over the knees of the seated person. Incidentally, the blanket **62** may extend to the chest of the seated person.

(31) A controller **100** is provided at a lower portion of the seat cushion **30**. A control system of the seat device **10** including the controller **100** will be described with reference to FIG. **2**.

(32) The controller **100** consists of an electronic information processing device including a microcomputer.

(33) To an input side of the controller **100**, an operation panel **70**, a breathing detecting device **72**, a heartbeat detecting device **74**, each body pressure detecting device **46**, a cabin temperature/humidity detecting device **76** (hereinafter abbreviated as “T/H detecting device **76**”), and a camera **78** are connected. The controller **100** acquires various pieces of information from these devices.

(34) The operation panel **70** consists of a touch panel or the like attached to a surface of a display device **98** such as a liquid crystal panel. The operation panel **70** is operated by an occupant so as to set an operation mode (automatic mode/manual mode) of the seat device **10**, switch on/off various functions, and perform time settings.

(35) The breathing detecting device **72** detects the breathing of the seated person by using the body pressure detecting devices **46**, a radio wave sensor (not shown), or the like. Also, the breathing detecting device **72** detects whether the breathing of the seated person is the abdominal breathing or the thoracic breathing. The breathing detecting device **72** may acquire breathing information on the seated person by communicating with a smartwatch or wearable terminal worn by the seated person. The breathing detecting device **72** may acquire the breathing information based on the video data by the camera **78**.

(36) The heartbeat detecting device **74** acquires heartbeat information on the seated person by communicating with the smartwatch or wearable terminal worn by the seated person.

(37) As described above, each body pressure detecting device **46** individually detects the body pressure of the seated person in each portion of the seat cushion **30** and the seat back **34**. The body pressure corresponds to the load of the seated person received by the seat cushion **30** and the seat back **34**.

(38) The T/H detecting device **76** detects the temperature and humidity inside the cabin around the seat device **10**.

(39) The camera **78** captures the video for observing the behavior of the seated person.

(40) Driving information and travel information on the automobile are input to the controller **100**. The driving information on the automobile includes information on whether the automobile is driven autonomously. The travel information on the automobile includes information on travel acceleration of the automobile.

(41) By executing a computer program, the controller **100** embodies a sleep-inducing breathing induction control unit **102** (hereinafter abbreviated as “sleep-inducing control unit **102**”), a sleeping level calculating unit **104**, a breathing induction control unit **106**, a seat temperature control unit **108**, a cabin environment control unit **110** (hereinafter abbreviated as “environment control unit **110**”), a massage control unit **112**, a posture control unit **114**, a reclining control unit **116**, and an exercise control unit **118**. Accordingly, the controller **100** performs various kinds of control.

(42) To an output side of the controller **100**, the back pressing device **40**, the waist pressing device **42**, each seat temperature adjusting device **52**, a cabin temperature/humidity adjusting device **90** (hereinafter abbreviated as “T/H adjusting device **90**”), each seat shape adjusting device **44**, the reclining device **32**, the ottoman **38**, the neck massaging device **48**, the calf massaging device **50**, a cabin sound device **92** (hereinafter abbreviated as “sound device **92**”), a cabin lighting device **94** (hereinafter abbreviated as “lighting device **94**”), a window shade device **96**, each knee back heater **56**, the warm air device **58**, and a display device **98** are connected. The controller **100** outputs signals such as control commands to these devices.

(43) The window shade device **96** includes a liquid crystal window shade or the like installed in the window pane **22**, and makes the window pane **22** opaque by an electric operation.
(44) The operation panel **70** and the display device **98** are arranged in front of the seat device **10** (for example, arranged in an instrument panel of the automobile) so as to be operated and seen by the seated person. The display device **98** displays a setting state and operation state of each operation of the seat device **10**. The setting state and operation state of each operation of the seat device **10** may be displayed on the smartwatch or wearable terminal of the seated person by using wireless communication.
(45) The sleep-inducing control unit **102** and the breathing induction control unit **106** control the back pressing device **40** and the waist pressing device **42**. The seat temperature control unit **108** controls the seat temperature adjusting device **52** and the knee back heater **56**. The environment control unit **110** controls the T/H adjusting device **90**, the sound device **92**, the lighting device **94**, and the warm air device **58**. The massage control unit **112** controls the seat shape adjusting device **44**, the neck massaging device **48**, and the calf massaging device **50**. The posture control unit **114** controls the seat shape adjusting device **44**. The reclining control unit **116** controls the reclining device **32** and the ottoman **38**. The exercise control unit **118** controls the seat shape adjusting device **44**.
(46) Next, each unit of the control system of the seat device **10** will be described in detail.
(47) When the operation mode of the seat device **10** set via the operation panel **70** is the automatic mode, the sleep-inducing control unit **102** is automatically activated as it is determined (detected) that the seated person is likely to sleep based on the information acquired by the breathing detecting device **72**, the heartbeat detecting device **74**, and the camera **78**. When the operation mode of the seat device **10** set via the operation panel **70** is the manual mode, the sleep-inducing control unit **102** is activated as a sleep induction mode is turned on according to an operation on the operation panel **70** by the occupant. In a case where the seat device **10** consists of the driver's seat, in any operation mode, the sleep-inducing control unit **102** can start only when the automobile is driven autonomously. In other words, in a case where the seat device **10** consists of the driver's seat, the sleep-inducing control unit **102** is prohibited from starting when the automobile is not driven autonomously.
(48) When turned on, the sleep-inducing control unit **102** controls the back pressing device **40** and the waist pressing device **42** such that the back pressing device **40** and the waist pressing device **42** press the back of the seated person at a set cycle (hereinafter referred to as “control target cycle”) corresponding to a breathing cycle of the seated person at a sleeping time. This control induces the breathing suitable for sleep induction.
(49) The sleep-inducing control unit **102** sots a default value of the control target cycle to a normal breathing cycle of a human at the sleeping time or at the time of the sleep induction. The sleep-inducing control unit **102** may modify the control target cycle upon learning the breathing cycle of the seated person detected by the breathing detecting device **72**. Accordingly, it is possible to properly induce sleep by inducing the breathing.
(50) The sleep-inducing control unit **102** determines whether the breathing of the seated person is the abdominal breathing or the thoracic breathing based on the information from the breathing detecting device **72** and the camera **78**. The sleep-inducing control unit **102** may perform control such that only one of the back pressing device **40** and the waist pressing device **42** operates so as to induce the breathing corresponding to a determination result thereof.
(51) Deep breathing according to the abdominal breathing makes the parasympathetic nerve dominant, and thus relaxes the mind and body. As the abdomen swells and deflates, the blood and lymph can smoothly circulate throughout the body, so that sleep is effectively induced. The thoracic breathing makes the sympathetic nerve dominant, and thus tensions the body and wakes up the seated person. Accordingly, the waist pressing device **42** may induce the abdominal breathing when sleep is induced, while the back pressing device **40** may induce the thoracic breathing when

the sleep ends (when the seated person wakes up). Accordingly, when sleep is induced, it is possible to induce the breathing suitable for inducing sleep. On the other hand, when sleep ends, it is possible to induce the breathing suitable for waking up the seated person.
(52) The sleep-inducing control unit **102** performs control such that the operation strength of the back pressing device **40** and the waist pressing device **42** is lowered as the sleeping level calculated by the sleeping level calculating unit **104** gets higher.
(53) As the back pressing device **40** and the waist pressing device **42** induce the breathing based on the above control, it is possible to better induce sleep and lead the seated person to sleep well.
(54) If the breathing of the seated person is unstable, the sleep-inducing control unit **102** controls the back pressing device **40** and the waist pressing device **42** so as to induce the breathing that eliminates unstableness in the breathing. As the breathing is induced by this control, it is possible to eliminate the unstableness in the breathing of the seated person.
(55) The sleep-inducing control unit **102** is kept on for a sleep time or nap time se by the seated person via the operation panel **70**. The sleep-inducing control unit **102** is automatically turned off after the sleep time or nap time elapses. Alternatively, the sleep-inducing control unit **102** is manually turned off according to an operation on the operation panel **70** by the seated person. The sleep-inducing control unit **102** may be linked with a smartwatch or a sleep meter at home to learn a recommended mode based on a sleeping state or health state of the seated person at night, so that the sleep time or nap time can be set individually.
(56) The sleeping level calculating unit **104** calculates the sleeping level of the seated person based on the information from the breathing detecting device **72**, the heartbeat detecting device **74**, the body pressure detecting device **46**, the camera **78**, and the like. Accordingly, it is possible to determine REM sleep and non-REM sleep relating to the depth of sleep.
(57) When the sleep-inducing control unit **102** is turned on, the seat temperature control unit **108**, the environment control unit **10**, the massage control unit **112**, the posture control unit **114**, and the reclining control unit **116** are turned on in a sleep-inducing mode so as to better induce sleep and lead the seated person to sleep well by the synergistic effect of these units and sleep-inducing breathing induction.
(58) In the sleep-inducing mode, that is, when the seated person is likely to sleep, the seat temperature control unit **108** slowly warms the seat surface so as to induce sleep of the seated person. When the sleeping level calculating unit **104** or the like determines that the seated person has fallen asleep, the seat temperature control unit **108** controls the seat temperature adjusting device **52** such that the seat temperature gets lower than the time before sleep. Hereinafter, the adjustment of the seat temperature by this control will be referred to as “sleep-inducing seat temperature adjustment”.
(59) Accordingly, it is possible to cause the effect of leading the seated person to sleep well according to the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42** and the sleep-inducing seat temperature adjustment.
(60) The environment control unit **110** refers to the cabin temperature/humidity detected by the T/H detecting device **76**, thereby controlling the T/H adjusting device **90** such that the cabin temperature and cabin humidity suitable for sleep are maintained in the sleep-inducing mode. For example, the environment control unit **110** controls the T/H adjusting device **90** such that the cabin temperature rises and the cabin humidity falls when the cabin is cooled, and the cabin temperature falls and the cabin humidity rises when the cabin is warmed.
(61) In the sleep-inducing mode (when the seated person is likely to sleep), the environment control unit **110** controls the sound device **92** such that the sound device **92** makes a sound with a rhythm suitable for the sleep induction and synchronized with the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42**, music with a rhythm that induces sleep, or quiet music suitable for inducing sleep.
(62) In the sleep-inducing mode (when the seated person is likely to sleep), the environment control

unit **110** controls the lighting device **94** and the window shade device **96** such that an inside of the cabin becomes dark and invisible from the outside.
(63) In the sleep-inducing mode (when the seated person is likely to sleep), the environment control unit **110** controls the warm air device **58** such that the warm air flows at the feet of the seated person.
(64) By the above control, a cabin environment (sleep-inducing environment) in which the seated person can easily sleep is organized. Accordingly, it is possible to cause the effect of leading the seated person to sleep by the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42** and the setting of the sleep-inducing environment.
(65) The environment control unit **110** can also perform control to change the brightness of the lighting device **94** and the sound of the sound device **92** according to the breathing rhythm of the seated person. This control makes it easier for the sealed person to be aware of his/her breathing, and thus it is possible to cause the effect of yoga or meditation.
(66) In the sleep-inducing mode (when the seated person is likely to sleep), the massage control unit **112** controls the seat shape adjusting device **44**, the neck massaging device **48** and the calf massaging device **50** to give a massage (sleep-inducing massage) suitable for the sleep induction and synchronized with the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42**. Accordingly, it is possible to cause the effect of leading the seated person to sleep well by the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42** and the sleep-inducing massage.
(67) In the sleep-inducing mode (when the seated person is likely to sleep), the posture control unit **114** controls the seat shape adjusting device **44** based on the body pressure distribution or the like detected by each body pressure detecting device **46** so as to correct the body pressure distribution, prevent bedsores, and prompt the seated person to roll over. In other words, the posture control unit **114** controls the seat shape adjusting device **44** such that a difference in the body pressure in each portion of the seat cushion **30** and the seat back **34** is reduced or the surfaces of the seat cushion **30** and the seat back **34** are deformed so as to prompt the seated person to change a posture.
(68) Accordingly, it is possible to cause the effect of leading the seated person to sleep well by performing the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42**, preventing bedsores, and prompting the seated person to roll over.
(69) In a case where the seated person sleeps while the automobile is traveling, the posture control unit **114** performs control such that the seat shape adjusting device **44** causes the seat cushion **30** and the seat back **34** to hold the seated person tight or hardens the seat cushion **30** and the seat back **34** according to the travel acceleration. Accordingly, it is possible to prevent the body of the seated person at the sleeping time from moving significantly due to the travel acceleration, and thus cause the effect of keeping stable sleep.
(70) In the sleep-inducing mode (when the seated person is likely to sleep), the reclining control unit **116** controls the reclining device **32** and the ottoman **38** so as to make the seat device **10** completely flat or nearly flat. Thus, the seated person can take a posture in which sleep is likely to be induced, so that the seated person can sleep well. Accordingly, it is possible to cause the effect of better inducing sleep and cause the seated person to sleep well by the above posture and the sleep-inducing breathing induction by the back pressing device **40** and the waist pressing device **42**.
(71) Due to the above functions and effects, the seated person is less likely to become uncomfortable at the sleeping time, and can sleep comfortably for a long time. Further, the seated person can be prevented from waking up earlier than the scheduled time when taking a nap.
(72) At the end of the sleep-inducing mode, the posture control unit **114** and the reclining control unit **116** perform, as a wake-up process, control to cause the seat shape adjusting device **44** to operate rapidly or vibrate, or to erect the seat back **34** by the reclining device **32**. This control prompts the sleeping seated person to wake up.

(73) If sleeping deeply at the time of short sleep such as a nap, the seated person may feel tired after waking up. As such, the sleeping level calculating unit **104** calculates the sleeping level. At the time of short sleep, the sleeping level calculating unit **104** may prompt the seated person to wake up at the stage where the sleeping level is low. At the time of long sleep, the sleeping level calculating unit **104** may prompt the seated person to wake up when the sleeping level is low (when REM sleep is caused). The seated person may be prompted to wake up as the seat shape adjusting device **44** operates rapidly or vibrates, the reclining device **32** erects the seat back **34**, the sound device **92** outputs a sound suitable for waking up the seated person, or the lighting device **94** lights the cabin suitably for waking up the seated occupant.
(74) Deep breathing according to the abdominal breathing makes the parasympathetic nerve dominant, and thus relaxes the mind and body. As the abdomen swells and deflates, the blood and lymph can smoothly circulate throughout the body, so that sleep is effectively induced. The thoracic breathing makes the sympathetic nerve dominant, and thus tensions the body and wakes up the seated person. Accordingly, the waist pressing device **42** may induce the abdominal breathing when sleep is induced, while the back pressing device **40** may induce the thoracic breathing when the sleep ends (when the seated person wakes up).
(75) The display device **98**, the smartphone of the seated person, or the tablet terminal of the seated person may display whether the breathing of the seated person is induced properly.
(76) The breathing induction control unit **106** is activated by the operation on the operation panel **70** by the seated person, and controls the back pressing device **40** and the waist pressing device **42** such that the breathing (the abdominal breathing or the thoracic breathing) of the seated person determined by the breathing detecting device **72** is induced. Accordingly, the breathing of the seated person that matches the current breathing state is induced, and thus the seated person can have a comfortable time with proper breathing. The breathing state of the seated person is displayed on the display device **98**.
(77) The breathing induction control unit **106** determines whether the seated person is excited based on the information from the breathing detecting device **72**, the heartbeat detecting device **74**, and the camera **78**. When the seated person is excited, the breathing induction control unit **106** controls the back pressing device **40** such that the thoracic breathing is induced. Accordingly, it is possible to prompt the seated person to calm down by breathing.
(78) The seat temperature control unit **108** is activated in a relaxing mode as the operation panel **70** is operated by the seated person. In the relaxing mode, the seat temperature control unit **108** controls the seat temperature adjusting device **52** such that the surface temperatures of the seat cushion **30** and the seat back **34** rise and fall repeatedly in a prescribed cycle. Accordingly, it is possible to relax the muscle of the seated person and thus cause a relaxing effect on the seated person.
(79) In this relaxing mode, the knee back heater **56** directly warms the knee back of the seated person and the warm air from the warm air device **58** warms the feet of the seated person.
(80) The massage control unit **112** is activated in the relaxing mode as the operation panel **70** is operated by the seated person. In the relaxing mode, the massage control unit **112** controls the seat shape adjusting device **44**, the neck massaging device **48**, and the calf massaging device **50** such that a massage to reduce the fatigue of the seated person is given. This massage may be given so as to match the rhythm of the breathing detected by the breathing detecting device **72**. Alternatively, this massage may be given with a rhythm that matches the timing of the breathing induction by the breathing induction control unit **106**.
(81) The massage by the seat shape adjusting device **44** may be given in a state where the seat temperature adjusting device **52** raises the surface temperatures of the seat cushion **30** and the seat back **34**. Further, the massage by the seat shape adjusting device **44** may be given strongly on a portion whose body pressure detected by the body pressure detecting device **46** is high so as to actively improve the blood flow in the portion whose body pressure is high.

(82) The seat shape adjusting device **44** can induce the breathing rhythm of the seated person by gradually slowing down the rhythm of the massage so as to prompt the seated person to breathe slowly and deeply. As the massage is given simultaneously with deep breathing, the relaxing effect can be enhanced. Further, this massage may be given at not only the seat cushion **30** and the seat back **34** but also an elbow rest (not shown) or the ottoman **38**.
(83) The posture control unit **114** is activated in the relaxing mode as the operation panel **70** is operated by the seated person. In the relaxing mode, the posture control unit **114** calculates the body pressure distribution based on the information from the body pressure detecting device **46**, and controls the seat shape adjusting device **44** such that the body pressure is evenly distributed. Accordingly, the body pressure is distributed in the seat cushion **30** and the seat back **34**, and thus it is possible to cause the effect of reducing the fatigue of the seated person seated for a long time.
(84) The reclining control unit **116** is activated as the operation panel **70** is operated by the seated person, and controls the reclining device **32** according to the operation on the operation panel **70**. Accordingly, a reclining state according to the request of the seated person is generated.
(85) The exercise control unit **118** is activated as the operation panel **70** is operated by the seated person, and controls the seat shape adjusting device **44** in an exercise mode. In the exercise mode, the seat shape adjusting device **44** moves synchronously with the breathing to cause the seated person to get some exercise of moving his/her waist synchronously with the breathing.
(86) Accordingly, the seated person gets some exercise of moving his/her waist synchronously with the breathing, so that a lack of exercise of the seated person can be effectively prevented, the health of the seated person can be effectively promoted, and a relaxing effect can be caused.
(87) The exercise control unit **118** controls the seat shape adjusting device **44** such that the rhythm of the exercise matches the rhythm of the breathing. Accordingly, it is possible to promote efficient breathing and improve an exercise effect. An exercise level of the exercise may correspond to the breathing and heart rate of the seated person. Accordingly, the seated person can get a reasonable exercise. At the end of the exercise, deep breathing may be prompted by the waist pressing device **42** or the like. In the exercise mode, the exercise control unit **118** may also cause a stretching effect to relax the body of the seated person.
(88) Next, the operation of the seat device **10** will be described with reference to a flowchart shown in FIG. **3**. The flowchart is performed by the controller **100**.
(89) First, the controller **100** determines whether the automobile is driven autonomously (step S**10**). If the automobile is driven autonomously (step S**10**: Yes), the controller **100** determines whether the operation mode of the seat device **10** is the automatic mode (step S**11**).
(90) If the operation mode is the automatic mode (step S**11**: Yes), the controller **100** makes a sleep-inducing determination (SI determination) (step S**12**). The sleep-inducing determination is a determination as to whether the seated person is likely to sleep. Upon determining based on various pieces of input information that the seated person is likely to sleep (step S**12**: Yes), the controller **100** executes the sleep-inducing breathing induction (SI induction) (step S**13**). Subsequently, the controller **100** executes the respective steps (step S**14** to step S**18**) of sleep-inducing seat temperature adjustment (SI adjustment), sleep-inducing reclination (SI reclination), a sleep-inducing massage (SI massage), a sleep-inducing environment setting (SI setting), and posture correction until the wake-up time arrives (step S**19**). As the initial setting of the automatic mode, at least one of the sleep-inducing seat temperature adjustment, the sleep-inducing reclination, the sleep-inducing massage, the sleep-inducing environment setting, and the posture correction may be selected via the operation panel **70** according to the preference of the seated person or the like.
(91) In the sleep-inducing breathing induction (step S**13**), the back pressing device **40** and the waist pressing device **42** apply a pressing stimulus suitable for the sleep induction to the back and waist of the seated person on a prescribed cycle. Accordingly, the breathing suitable for the sleep induction is induced.
(92) In the sleep-inducing seat temperature adjustment (step S**14**) under the sleep-inducing

breathing induction, the seat temperature adjusting device **52** slowly warms the surface of the seat to induce sleep of the seated person. When the seated person falls asleep, the seat temperature adjusting device **52** makes the seat temperature lower than the time before sleep, thereby generating a comfortable sleeping environment.
(93) In the sleep-inducing reclination (step S**15**) under the sleep-inducing breathing induction, the reclining device **32** and the ottoman **38** make the seat device **10** completely flat or nearly flat, thereby generating the comfortable sleeping environment.
(94) In the sleep-inducing massage (step S**16**) under the sleep-inducing breathing induction, the seat shape adjusting device **44**, the neck massaging device **48**, and the calf massaging device **50** give a massage synchronized with the sleep-inducing breathing induction.
(95) In the sleep-inducing environment setting (step S**17**), the sound device **92** outputs a sound suitable for the sleep induction, the lighting device **94** lights the cabin suitably for the sleep induction, and the window shade device **96** shields the window pane **22** suitably for the sleep induction.
(96) In the posture correction (step S**18**), a posture correction routine shown in FIG. **4** is performed.
(97) The posture correction routine shown in FIG. **4** will be described.
(98) First, each body pressure detecting device **46** detects the body pressure in each portion of the seat cushion **30** and the seat back **34**, thereby acquiring the body pressure information (step S**40**). Next, the controller **100** calculates the body pressure distribution (BPD) in the seat cushion **30** and the seat back **34** based on the body pressure information on each portion (step S**41**), and determines whether the body pressure distribution is appropriate (step S**42**). One of the conditions for the body pressure distribution to be appropriate is that there is no portion where the body pressure is locally higher than the other portions.
(99) If the body pressure distribution is not appropriate (step S**42**: No), the seat shape adjusting device **44** performs a correction process for adjusting the body pressure distribution (step S**43**).
(100) If the body pressure distribution is appropriate (step S**42**: Yes), the controller **100** determines whether to prompt the seated person to roll over (step S**44**). Determination as to whether to prompt the seated person to roll over is made based on the breathing, heartbeat, video data of the seated person, and the like. Upon determining to prompt the seated person to roll over (step S**44**: Yes), the seat shape adjusting device **44** changes the surface shapes of the seat cushion **30** and the seat back **34** into a shape that prompts the seated person to roll over or a shape that makes it easy for the seated person to roll over (step S**45**).
(101) Next, the travel acceleration (hereinafter abbreviated as “acceleration”) of the automobile is detected (step S**46**), and the controller **100** determines whether the acceleration is equal to or greater than a prescribed value (step S**47**). When the acceleration is equal to or greater than the prescribed value (step S**47**: Yes), the seat shape adjusting device **44** causes the seat cushion **30** and the seat back **34** to hold the seated person tight and hardens the seat cushion **30** and the seat back **34** (step S**48**). Accordingly, the posture correction routine ends.
(102) Next, returning to the flowchart shown in FIG. **3**, the controller **100** determines whether a prescribed wake-up time arrives (step S**19**) after the above-mentioned steps S**13** to S**18**. When the wake-up time arrives (step S**19**: Yes), the controller **100** executes a wake-up process step S**20**).
(103) In the sleep-inducing determination in step S**12**, upon determining that the seated person is not likely to sleep (step S**12**: No), the controller **100** determines whether the seated person is excited (step S**21**). This determination is made based on the change in the breathing cycle or heartbeat of the seated person, or the video data of the seated person. Upon determining that the seated person is excited (step S**21**: Yes), the controller **100** performs an assuaging process of the seated person, namely, a process of suppressing the excitement of the seated person (step S**22**). The assuaging process includes induction of the abdominal breathing by the waist pressing device **42**, a massage by the seat shape adjusting device **44**, the neck massaging device **48**, and the calf massaging device **50** to improve relaxation.

(104) In the determination of the driving mode of the automobile in step S**10**, upon determining that the automobile is not driven autonomously, that is, upon determining that the automobile is manually driven by the driver (step S**10**: No), the controller **100** determines whether the sleep-inducing mode (SI mode) is selected (step S**23**). Also, in the determination of the operation mode of the seat device **10** in step S**11**, upon determining that the operation mode is not the automatic mode, that is, upon determining that the operation mode is the manual mode (step S**11**: No), the controller **100** determines whether the sleep-inducing mode (SI mode) is selected (step S**23**).
(105) When the sleep-inducing mode is selected (step S**23**: Yes), the controller **100** proceeds to step S**13**, and sequentially executes steps S**13** to S**20** in the same manner as the automatic mode.
(106) In a case where the sleep-inducing mode is not selected. (step S**23**: No), the controller **100** determines whether the seated person is excited (step S**24**). This determination in step S**24** is the same as the determination in step S**21**. Upon determining that the seated person is excited (step S**24**: Yes), the controller **100** executes an assuaging process of the seated person like step S**22** (step S**25**).
(107) Thereafter, the controller **100** determines whether a warming/cooling mode (w/c mode) is selected (step S**26**). When the warming/cooling mode is selected (step S**26**: Yes), the controller **100** executes a warming/cooling routine (w/c routine) shown in FIG. **5** (S**27**).
(108) The warming/cooling routine (w/c routine) shown in FIG. **5** will be described.
(109) First, the controller **100** starts a cooling mode by the Peltier element or the like of the seat temperature adjusting device **52** (step S**50**), and monitors whether the surface temperature of the seat (the seat cushion **30** and the seat back **34**) has fallen to a prescribed value (step S**51**). When the surface temperature of the seat has fallen to the prescribed value (step S**51**: Yes), the controller **100** maintains this temperature state for a prescribed period (step S**52**).
(110) When the prescribed temperature state is maintained for the prescribed period, the controller **100** ends the cooling mode (step S**53**), starts a warming mode by the heater of the seat temperature adjusting device **52** (step S**54**), and monitors whether the surface temperature of the seat has risen to a prescribed value (step S**55**). When the surface temperature of the seat has risen to the prescribed value (step S**55**: Yes), the controller **100** maintains this temperature state for a prescribed period (step S**56**). When the prescribed temperature state is maintained for the prescribed period, the controller **100** ends the warming mode (step S**57**), and increments the counter of the number of times (step S**58**).
(111) The controller **100** repeatedly executes steps S**50** to S**58** until reaching prescribed times. Upon reaching the prescribed times (step S**59**: Yes), the controller **100** ends the warming cooling routine.
(112) Next, returning to the flowchart shown in FIG. **3**, the controller **100** determines whether a massage mode is selected (step S**28**). When the massage mode is selected (step S**28**: Yes), the controller **100** performs a massage routine shown in FIG. **6** (S**29**).
(113) The massage routine shown in FIG. **6** will be described.
(114) First, prior to the massage, the heater of the seat temperature adjusting device **52** starts heating the seat to raise the surface temperature thereof (step S**60**). The controller **100** monitors whether a prescribed temperature is maintained for a prescribed period (step S**61**). When the prescribed temperature is maintained for the prescribed period (step S**61**: Yes), the controller **100** determines an initial rhythm and final rhythm of the massage (step S**62**). The initial rhythm and final rhythm of the massage may be determined by the seated person according to the degree of the massage desired by the seated person.
(115) Next, in a state where the breathing of the seated person is detected (step S**63**), the seat shape adjusting device **44** starts the massage with the initial rhythm (step S**64**). If a prescribed period elapses since the start of the massage (step S**65**: Yes), the seat shape adjusting device **44** slowly changes the rhythm of the massage (step S**66**) such that the rhythm of the massage approaches the final rhythm, and the controller **100** monitors whether the rhythm of the massage matches the

rhythm of the breathing (step S**67**).

(116) When the rhythm of massage matches the rhythm of the breathing (step S**67**: Yes), the controller **100** determines whether the current massage rhythm is the final rhythm (step S**68**). In a case where the current massage rhythm is not the final rhythm (step S**68**: No), the controller **100** repeats steps S**66** and S**67**. In a case where the current massage rhythm is the final rhythm (step S**68**: Yes), the seat shape adjusting device **44** maintains the final rhythm until a prescribed period elapses (step S**69**). After that, the controller **100** ends the massage routine.

(117) Next, returning to the flowchart shown in FIG. **3**, the controller **100** determines whether the exercise mode is selected (step S**30**). When the exercise mode is selected (step S**30**: Yes), the seat shape adjusting device **44** causes the seated person to exercise (step S**31**).

(118) Next, the controller **100** determines whether a breathing induction mode (BI mode) is selected (step S**32**). In a case where the breathing induction mode is selected (step S**32**: Yes), the controller **100** performs a breathing induction routine (BI routine) shown in FIG. **7** (S**33**).

(119) The breathing induction routine (BI routine) shown in FIG. **7** will be described.

(120) First, the current breathing state is detected (step S**70**), and the controller **100** determines whether the thoracic breathing (TB) is appropriate based on the current breathing condition (step S**71**). In a case where the thoracic breathing is appropriate (step S**71**: Yes), the back pressing device **40** induces the thoracic breathing and the fact that the thoracic breathing is induced is displayed (step S**72**). After that, the current breathing state is detected (step S**73**), and the controller **100** determines whether the thoracic breathing is performed (step S**74**). In a case where the thoracic breathing is performed (step S**74**: Yes), the controller **100** returns to step S**73**. In a case where the thoracic breathing is not performed (step S**74**: No), the controller **100** notifies the seated person that the thoracic breathing is not performed appropriately (step S**75**), and returns to step S**73**.

(121) In the determination of step S**71**, in a case where the controller **100** determines that the thoracic breathing is not appropriate (step S**71**: No), that is, in a case where the controller **100** determines that the abdominal breathing (AB) is appropriate, the waist pressing device **42** induces the abdominal breathing and the fact that the abdominal breathing is induced is displayed (step S**76**). After that, the current breathing state is detected (step S**77**), and the controller **100** determines whether the abdominal breathing is performed (step S**78**). In a case where the abdominal breathing is performed (step S**78**: Yes), the controller **100** returns to step S**77**. In a case where the abdominal breathing is not performed (step S**78**: No), the controller **100** notifies the seated person that the abdominal breathing is not performed appropriately (step S**79**), and returns to step S**77**. When the selection of the breathing induction mode is canceled, the breathing induction routine ends.

(122) Next, one embodiment of a power supply mechanism that supplies electric power from a power supply (not shown) on a vehicle body side to each portion of the seat device **10** will be described with reference to FIGS. **8** and **9**.

(123) The seat cushion **30** includes a seat cushion frame **120** having a substantially square frame-like shape. The seat back **34** includes a seat back frame **122** having a substantially square frame-like shape.

(124) A power transmitter **132** is attached to an upper surface of the lower rail **16**. The power transmitter **132** includes a band-shaped power transmitting electrode **130** extending over substantially the entire length of the lower rail **16**. A power receiver **136** including a power receiving electrode **134** is attached to a lower surface of the upper rail **18**. The power transmitting electrode **130** and the power receiving electrode **134** face each other with a prescribed gap therebetween, and compose a non-contact type power supply mechanism using electric field coupling. Accordingly, the electric power is supplied from the power supply (not shown) on the vehicle body side to each portion of the seat device **10**.

(125) A band-shaped bellows **138** connected to the upper rail **18** is attached to the lower rail **16** so as to close an upper opening of the lower rail **16**. The bellows **138** protects the power transmitting electrode **130**.

(126) In another embodiment, as shown in FIG. **10**, the power transmitting electrode **130** and the power receiving electrode **134** have uneven surfaces that are opposed to each other so as to increase opposed areas of the power transmitting electrode **130** and the power receiving electrode **134**. As the opposed areas increase, the transmittable electric power increases.
(127) FIG. **11** shows a power supply mechanism according to still another embodiment. In this embodiment, the power transmitting electrode **130** and the power receiving electrode **134** extend between the left and right lower rails **16**. Accordingly, the opposed areas of the power transmitting electrode **130** and the power receiving electrode **134** increase, and thus the transmittable electric power increases.
(128) Incidentally, in this embodiment, the power transmitting electrode **130** and the power receiving electrode **134** may have uneven surfaces that are opposed to each other, like the embodiment shown in FIG. **10**.
(129) FIG. **12** shows a power supply mechanism according to still another embodiment. In this embodiment, the lower rail **16** has a box-shaped section **17**. The power transmitting electrode **130** is provided on an inner surface of the box-shaped section **17**. The upper rail **18** includes a slider **19** that enters the box-shaped section **17** from the inside. The power receiving electrode **134** is provided on an outer surface of the slider **19**. In this embodiment, the power transmitting electrode **130** is not exposed to the outside.
(130) Next, a power supply mechanism according to still another embodiment will be described with reference to FIGS. **13** and **14**. The power supply mechanism supplies the electric power from a power supply (not shown) on a vehicle body side to each portion of the seat device **10**.
(131) A power transmitter **150** is attached to the floor panel **14** between the left and right lower rails **16**. Beams **128** extending in the front-and-rear direction is attached to a front member **124** and a rear member **126** of the seat cushion frame **120**. A power receiver **154** is suspended from and supported by the beams **128** via dampers **152**.
(132) The power transmitter **150** and the power receiver **154** are opposed to each other with a prescribed gap therebetween, and compose a non-contact type power supply mechanism using magnetic field resonance. Accordingly, the electric power is supplied from the power supply (not shown) on the vehicle body side to each portion of the seat device **10**.
(133) A magnetic shield member **156** is attached to the beams **128** so as to surround the power receiver **154**.
(134) In this embodiment, the dampers **152** suppress the transmission of vibration from the side of the seat cushion frame **120** to the power receiver **154**.
(135) FIG. **15** shows a power supply mechanism using the magnetic field resonance according to still another embodiment. On inner surfaces of the left and right lower rails **16**, a plurality of power transmitters **150** is provided at prescribed intervals along an extending direction of the lower rails **16**.
(136) Preferable embodiments of the present invention have been described in the foregoing. However, as easily understood by a skilled person, the present invention should not be limited by the foregoing embodiments, and various modifications and alterations are possible within the scope of the present invention. For example, the back pressing device **40**, the waist pressing device **42**, and the seat shape adjusting device **44** are not limited to an air pad, and may consist of a mechanical device using an electrostriction element that is deformed by the electric power, a cam, or the like. The exercise may not be realized by the seat shape adjusting device **44** but by a known electric lumbar support device installed in the seat back **34**.
(137) Also, not all components shown in the foregoing embodiment are necessarily indispensable, and they may be selectively used as appropriate within the scope of the present invention.
Glossary of Terms
(138) **10**: seat device **12**: slide rail device **14**: floor panel **15**: bracket **16**: lower rail **17**: box-shaped section **18**: upper rail **19**: slider **20**: side door **22**: window pane **30**: seat cushion **32**: electric

reclining device **34**: seat back **36**: headrest **38**: electric ottoman **40**: back pressing device **42**: waist pressing device **44**: seat shape adjusting device (shape adjusting device) **46**: body pressure detecting device **48**: neck massaging device **50**: calf massaging device **52**: seat temperature adjusting device (temperature adjusting device **56**: knee back heater **58**: warm air device **60**: blanket box **62**: blanket **64**: locking hole **66**: engagement hook **70**: operation panel **72**: breathing detecting device **74**: heartbeat detecting device **76**: cabin temperature/humidity detecting device **78**: camera **90**: cabin temperature/humidity adjusting device **92**: cabin sound device **94**: cabin lighting device **96**: window shade device **98**: display device **100**: controller **102**: sleep-inducing breathing control unit **104**: sleeping level calculating unit **106**: breathing control unit **108**: seat temperature control unit **110**: cabin environment control unit **112**: massage control unit **114**: posture control unit **116**: reclining control unit **118**: exercise control unit **120**: seat cushion frame **122**: seat back frame **124**: front member **126**: rear member **128**: beam **130**: power transmitting electrode **132**: power transmitter **134**: power receiving electrode **136**: power receiver **138**: bellows **150**: power transmitter **152**: damper **154**: power receiver **156**: magnetic shield member

Claims

1. A seat device, comprising: a seat body including a seat cushion and a seat back; a pressing device provided in the seat back and configured to press a back or a waist of a seated person so as to induce breathing thereof; shape adjusting devices configured to change a surface shape of the seat back; and a controller configured to control the pressing device such that the pressing device presses the back or the waist of the seated person at a set cycle corresponding to a breathing cycle of a person at a sleeping time, and control the shape adjusting devices, wherein the controller is configured to determine whether the breathing of the seated person is abdominal breathing or thoracic breathing, and control the pressing device so as to induce the abdominal breathing or the thoracic breathing corresponding to a determination result of whether the breathing of the seated person is the abdominal breathing or the thoracic breathing, wherein the pressing device includes: a back pressing device configured to press the back of the seated person so as to induce the thoracic breathing; and a waist pressing device configured to press the waist of the seated person to induce the abdominal breathing, and wherein the controller is configured to cause the waist pressing device to operate so as to induce the abdominal breathing when sleep is induced, and cause the back pressing device to operate so as to induce the thoracic breathing when the sleep ends, wherein the waist pressing device is arranged below the back pressing device, and the shape adjusting devices are provided in the seat back, arranged between the waist pressing device and the back pressing device, and spaced away from each other in a lateral direction, wherein the seat device further comprises a plurality of body pressure detecting devices provided in the seat back and configured to detect a body pressure of the seated person, the plurality of body pressure detecting devices includes: a plurality of first body pressure detecting devices arranged between the shape adjusting devices and the waist pressing device and spaced away from each other in the lateral direction; and a plurality of second body pressure detecting devices arranged below the waist pressing device and spaced away from each other in the lateral direction, and each of the shape adjusting devices, each of the first body pressure detecting devices, and each of the second body pressure detecting devices are aligned in an up-and-down direction.
2. The seat device according to claim 1, wherein the controller is configured to correct the set cycle upon learning the breathing cycle of the seated person at the sleeping time.
3. The seat device according to claim 1, further comprising a breathing detecting device configured to detect the breathing of the seated person, wherein in a case where the breathing detected by the breathing detecting device is unstable, the controller controls the pressing device so as to induce the breathing that eliminates unstableness in the breathing.
4. The seat device according to claim 1, further comprising a warm air device configured to warm

feet of the seated person, wherein the controller is configured to determine based on external information whether the seated person is likely to sleep, and perform control to turn on the pressing device and the warm air device upon determining that the seated person is likely to sleep.
5. The seat device according to claim 1, wherein the controller is configured to determine, based on external information, whether the seated person is likely to sleep, and turn on the pressing device and control the shape adjusting devices such that a massage synchronized with a breathing rhythm of the seated person is given upon determining that the seated person is likely to sleep.
6. The seat device according to claim 1, wherein the controller is configured to control the shape adjusting devices such that a massage synchronized with a breathing rhythm of the seated person is given.
7. The seat device according to claim 1, wherein the controller is configured to control the shape adjusting devices such that a difference in the body pressure in each portion is reduced.
8. The seat device according to claim 1, wherein the controller is configured to control the shape adjusting devices such that surfaces of the seat cushion and the seat back are deformed so as to prompt the seated person to change a posture.
9. The seat device according to claim 1, wherein the controller is configured to control the shape adjusting devices such that surfaces of the seat cushion and the seat back are deformed at a set wake-up time so as to promote the seated person to wake up.
10. The seat device according to claim 1, wherein the seat device is for a vehicle.
11. The seat device according to claim 1, wherein the seat device is for an automobile configured to be driven autonomously.
12. The seat device according to claim 1, wherein the controller is configured to cause only one of the back pressing device and the waist pressing device to operate so as to induce the breathing corresponding to the determination result thereof.
13. The seat device according to claim 1, wherein the controller includes a posture control unit configured to control the shape adjusting devices.
14. The seat device according to claim 13, wherein in a sleep-inducing mode for inducing sleep of the seated person, the posture control unit controls the shape adjusting devices based on distribution of the body pressure detected by each of the body pressure detecting devices so as to correct the distribution of the body pressure, prevent a bedsore, or prompt the seated person to roll over.
15. The seat device according to claim 13, wherein the seat device is provided in an automobile, and in a case where the seated person sleeps while the automobile is traveling, the posture control unit performs control such that the shape adjusting devices cause the seat cushion and the seat back to hold the seated person tight or harden the seat cushion and the seat back according to a travel acceleration.
16. A seat device, comprising: a seat body including a seat cushion and a seat back; a pressing device provided in the seat back and configured to press a back or a waist of a seated person so as to induce breathing thereof; shape adjusting devices configured to change a surface shape of the seat back; and a controller configured to control the pressing device such that the pressing device presses the back or the waist of the seated person at a set cycle corresponding to a breathing cycle of a person at a sleeping time, and control the shape adjusting devices, wherein the controller is configured to control the pressing device so as to induce abdominal breathing when sleep is induced, and induce thoracic breathing when the sleep ends, wherein the pressing device includes: a back pressing device configured to press the back of the seated person so as to induce the thoracic breathing; and a waist pressing device configured to press the waist of the seated person to induce the abdominal breathing, and wherein the controller is configured to cause the waist pressing device to operate so as to induce the abdominal breathing when the sleep is induced, and cause the back pressing device to operate so as to induce the thoracic breathing when the sleep ends, wherein the waist pressing device is arranged below the back pressing device, and the shape adjusting

devices are provided in the seat back, arranged between the waist pressing device and the back pressing device, and spaced away from each other in a lateral direction, wherein the seat device further comprises a plurality of body pressure detecting devices provided in the seat back and configured to detect a body pressure of the seated person, the plurality of body pressure detecting devices includes: a plurality of first body pressure detecting devices arranged between the shape adjusting devices and the waist pressing device and spaced away from each other in the lateral direction; and a plurality of second body pressure detecting devices arranged below the waist pressing device and spaced away from each other in the lateral direction, and each of the shape adjusting devices, each of the first body pressure detecting devices, and each of the second body pressure detecting devices are aligned in an up-and-down direction.

17. The seat device according to claim 16, wherein the controller includes a posture control unit configured to control the shape adjusting devices.

18. The seat device according to claim 17, wherein in a sleep-inducing mode for inducing sleep of the seated person, the posture control unit controls the shape adjusting devices based on distribution of the body pressure detected by each of the body pressure detecting devices so as to correct the distribution of the body pressure, prevent a bedsore, or prompt the seated person to roll over.

19. The seat device according to claim 18, wherein the seat device is provided in an automobile, and in a case where the seated person sleeps while the automobile is traveling, the posture control unit performs control such that the shape adjusting devices cause the seat cushion and the seat back to hold the seated person tight or harden the seat cushion and the seat back according to a travel acceleration.
